NERVOUS SYSTEM
DIVISIONS CNS, PNS 2

LEARNING OBJECTIVES

At the end of the lecture the student should be able to;

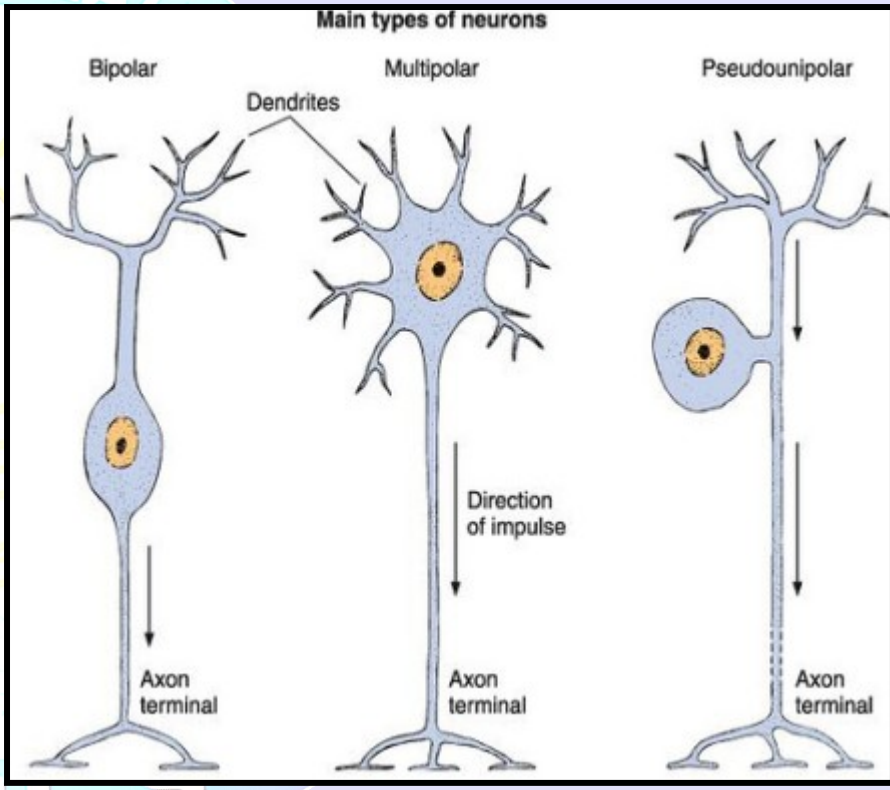
- Discusses the classification of neurons on the basis of
  - Number of processes
  - Length of fibers

- Define a nerve and its coverings
-
- Differentiate between myelinated and unmyelinated fibres

- Enlist various types of Neuroglia and state their functions

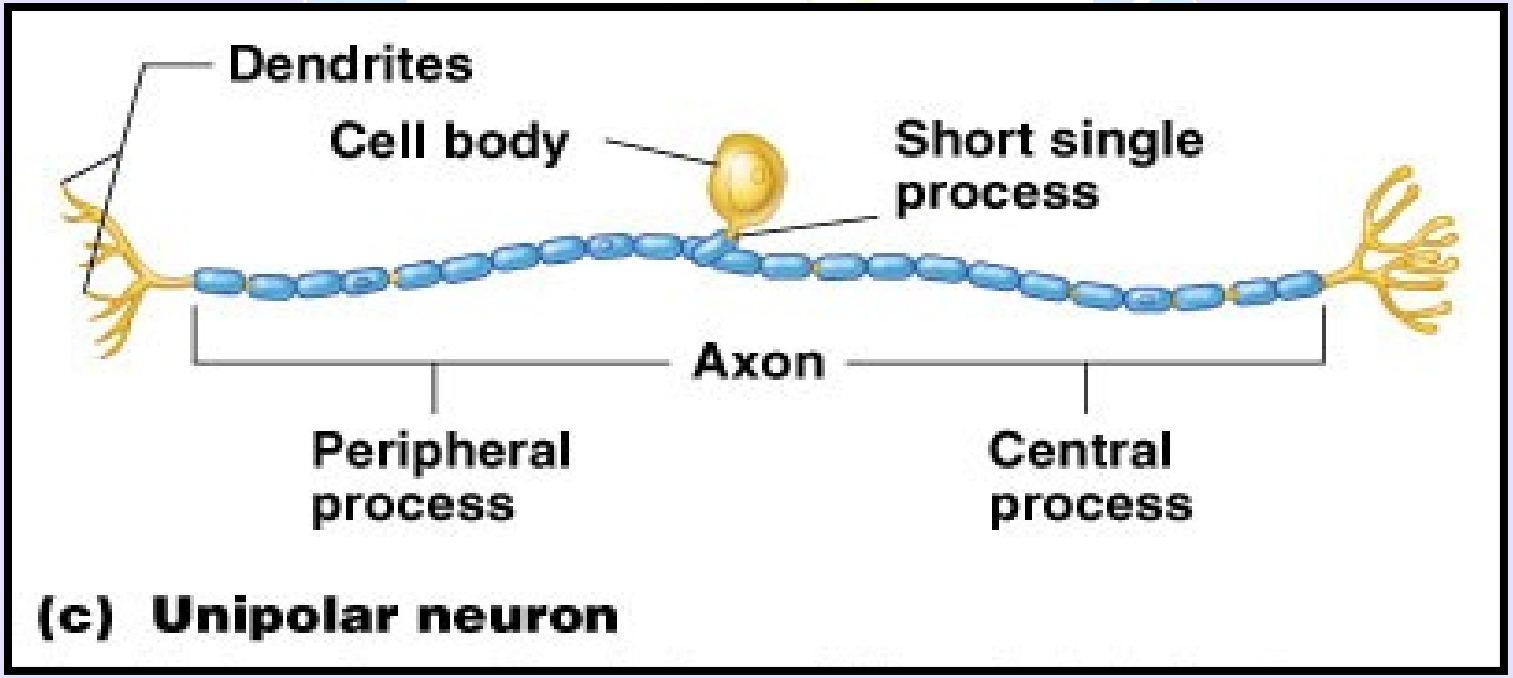
STRUCTURAL CLASSIFICATION OF NEURON

- **Unipolar:**
- Dendrite and axon emerging from same process
- **Pseudounipolar:**
  - Single process arises from body
  - Branches into an axon and dendrite
  - e.g., Present in spinal and cranial ganglia (sensory neuron)
- **Bipolar:**
  - Single axon and single dendrite on opposite ends of the soma. e.g., interneuron
- **Multipolar;**
  - Single axon & multiple dendrites
  - Most common type in men
  - e.g., Motor cortex
- **Golgi I**: neurons with long-projecting axonal processes.
- **Golgi II**: neurons whose axonal process projects locally

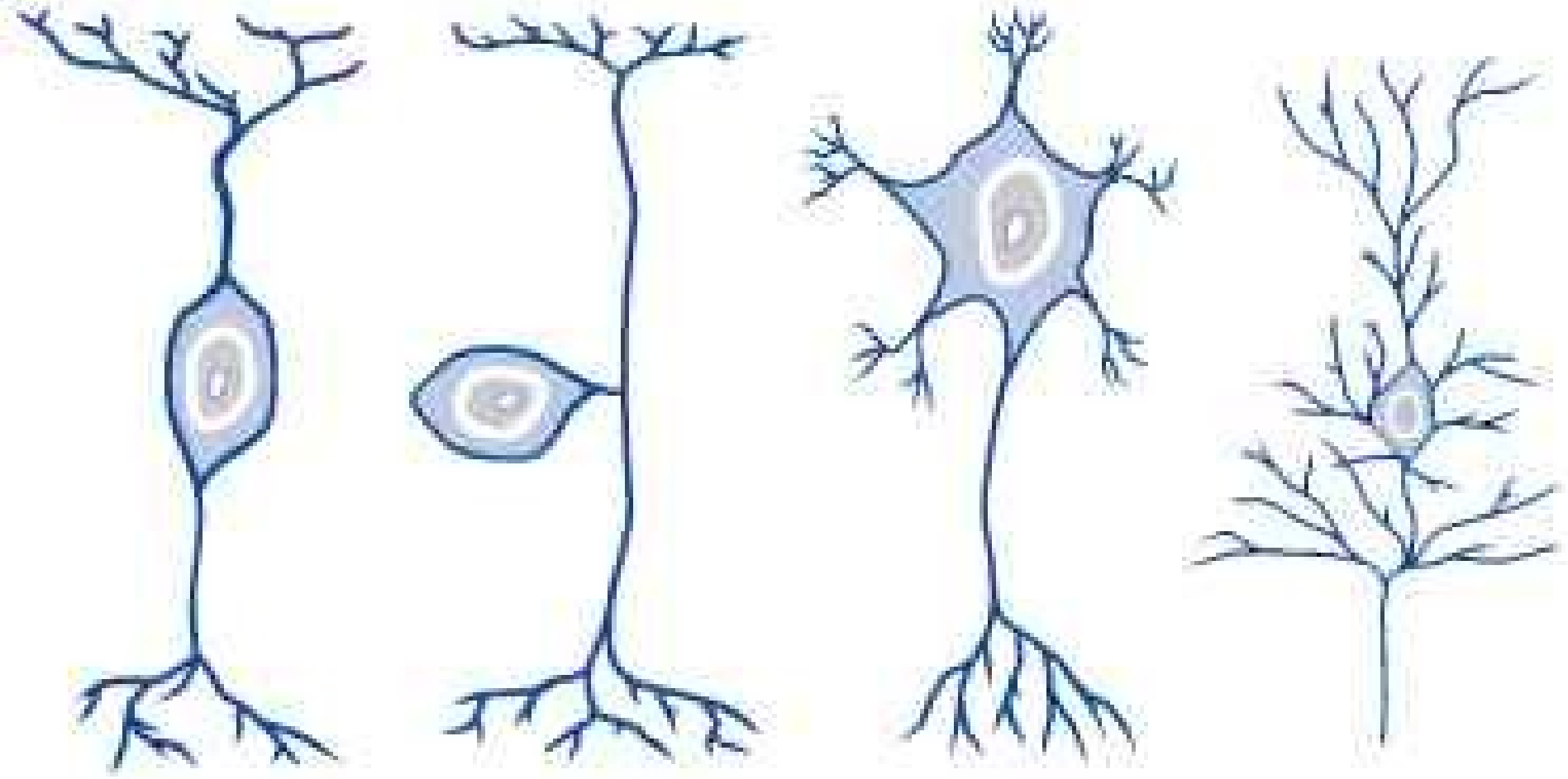


CLASSIFICATION OF NEURONS

- Unipolar neurons – have a short single process leaving the cell body



Basic Neuron Types



Bipolar (Interneuron)

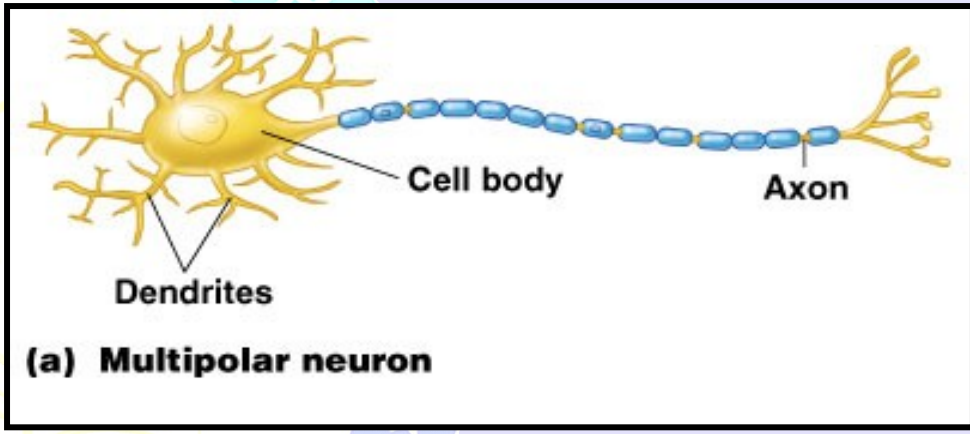
Unipolar (Sensory Neuron)

Multipolar (Motoneuron)

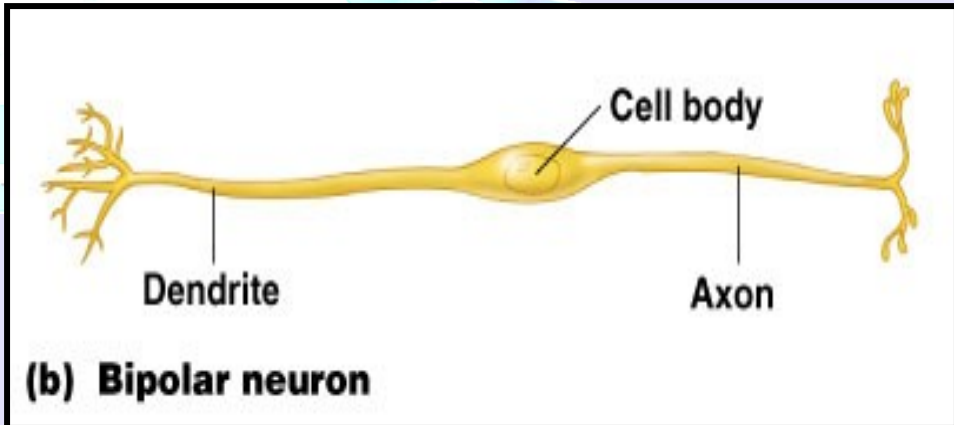
Pyrimidal Cell

Classification of Neurons

- Multipolar neurons – many extensions from the cell body

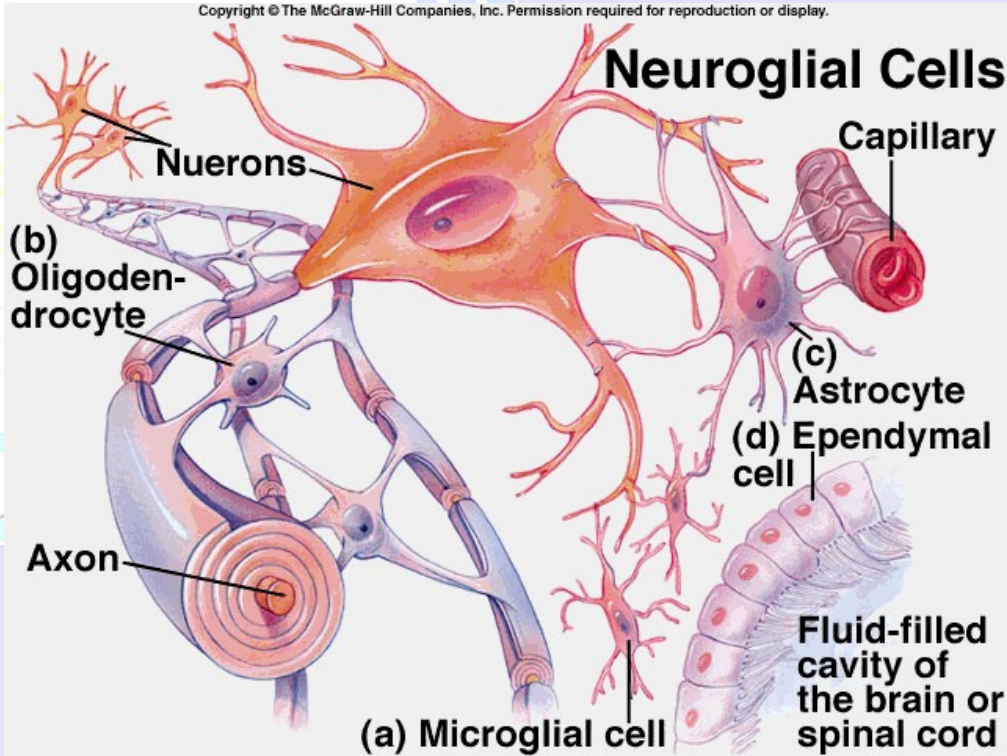


- Bipolar neurons – one axon and one dendrite



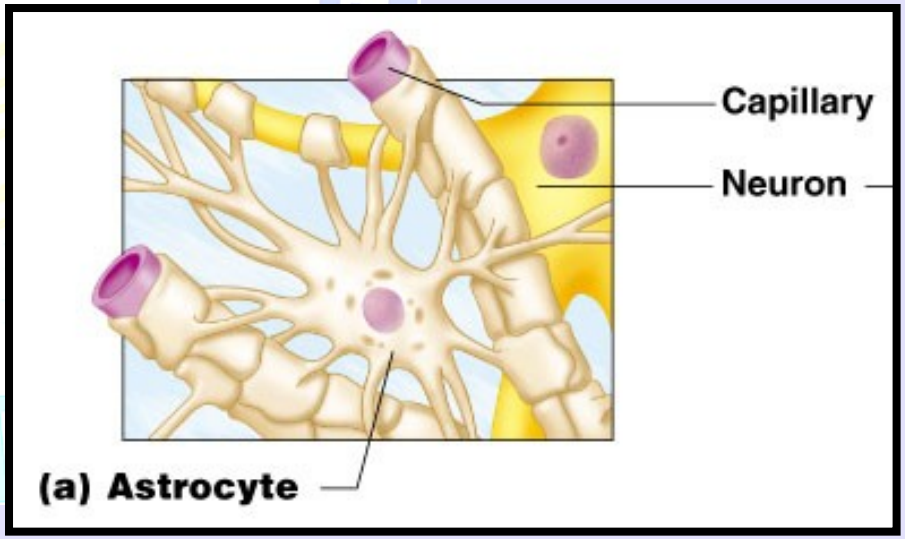
NEUROGLIA

- **Neuroglia** is the supporting or glial tissue that is 10-times more abundant in mammalian brain than neurons



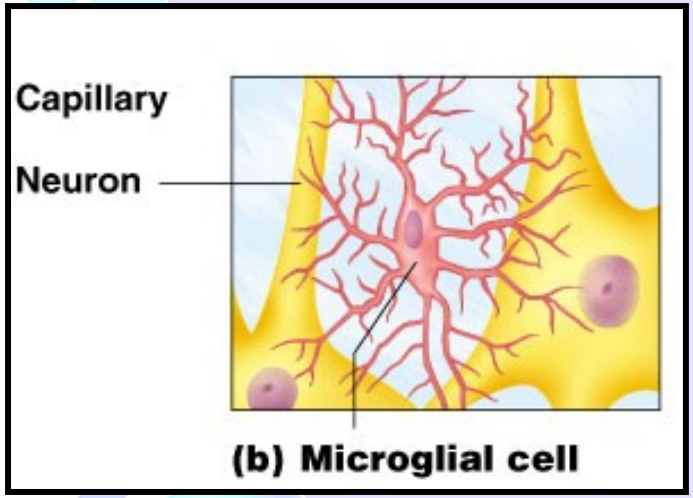
NERVOUS TISSUE: SUPPORT CELLS (NEUROGLIA OR GLIA)

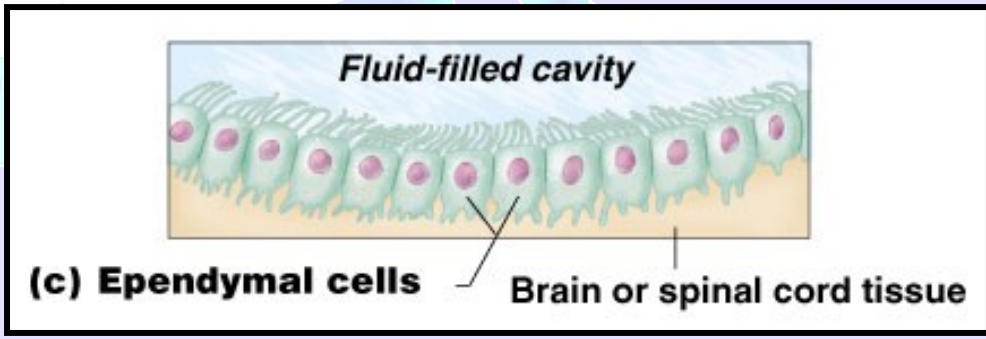
- **Astrocytes**
  - Abundant, star-shaped cells
  - Form barrier between capillaries and neurons
  - Control the chemical environment of the brain (CNS)



NERVOUS TISSUE: SUPPORT CELLS

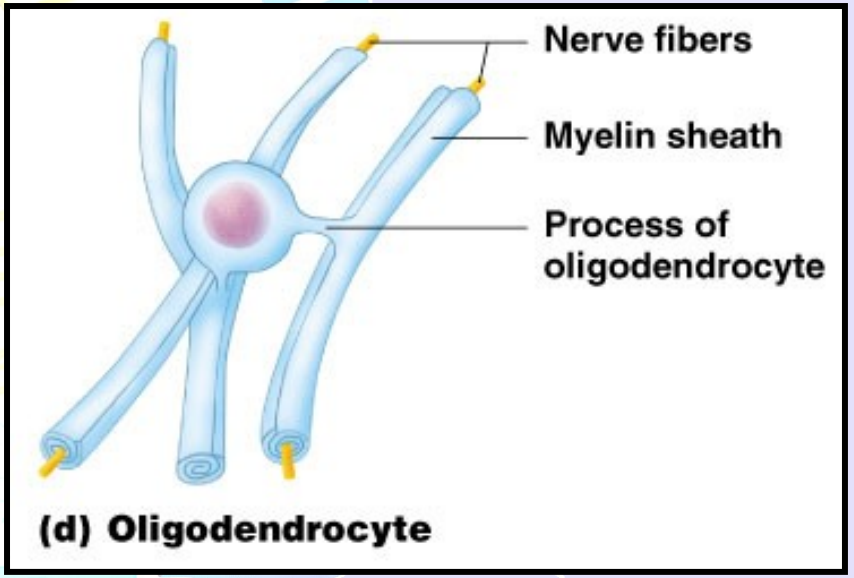
- **Microglia (CNS)**
  - Spider-like phagocytes
  - Dispose of debris
- **Ependymal cells (CNS)**
  - Line cavities of the brain and spinal cord
  - Circulate cerebrospinal fluid





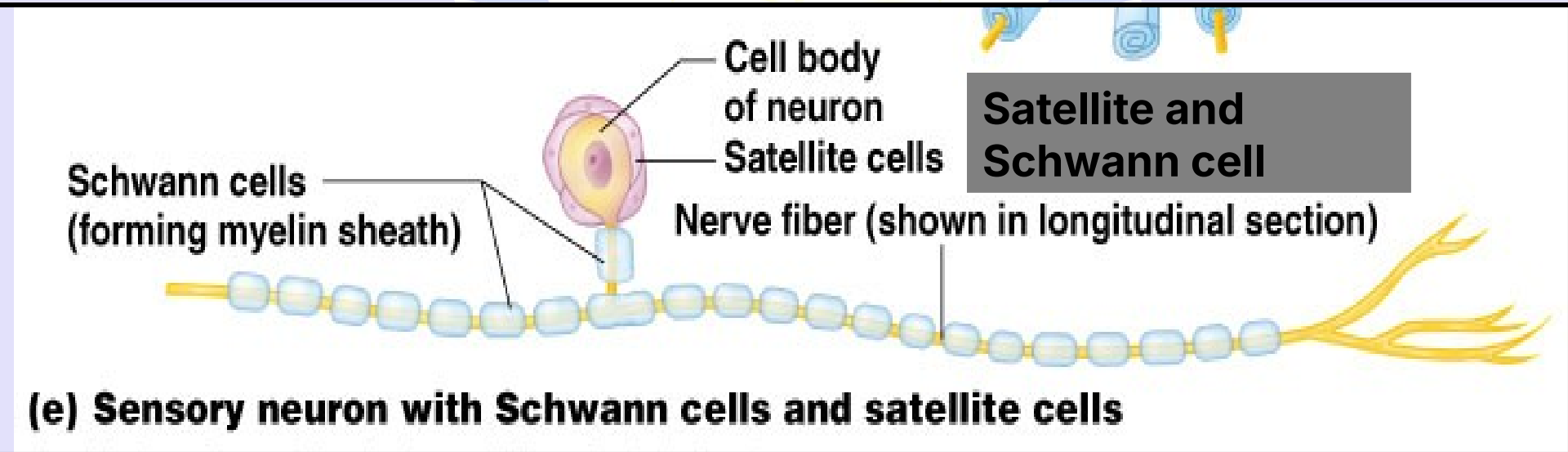
NERVOUS TISSUE: SUPPORT CELLS

- **Oligodendrocytes (CNS)**
  - Produce myelin sheath around nerve fibers in the central nervous system

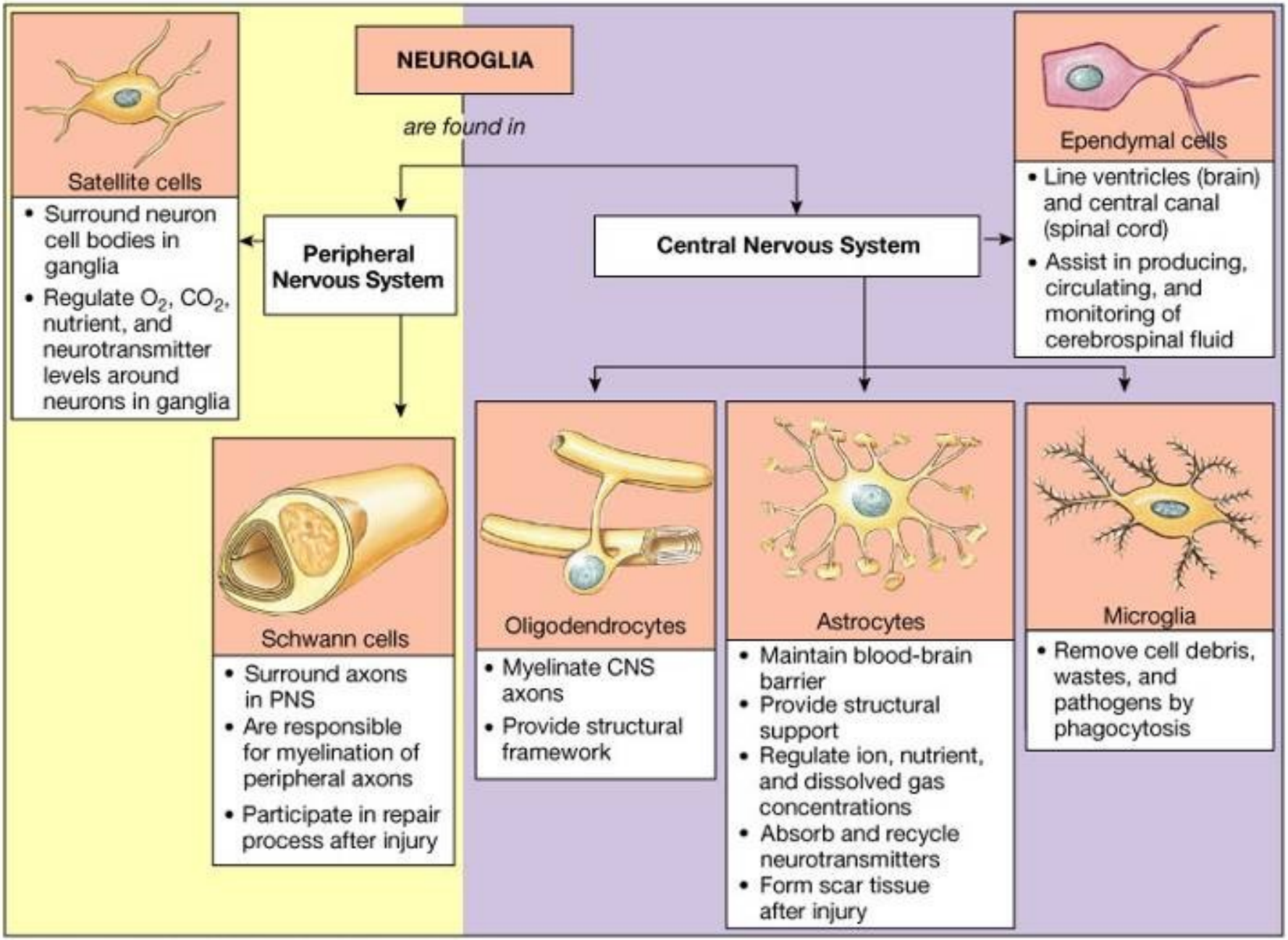


SUPPORT CELLS OF THE PNS

- **Satellite cells**
  - Protect neuron cell bodies
- **Schwann cells**
  - Form myelin sheath in the peripheral nervous system



(e) Sensory neuron with Schwann cells and satellite cells


NEUROGLIA
are found in
Satellite cells
• Surround neuron cell bodies in ganglia
• Regulate O_2, CO_2, nutrient, and neurotransmitter levels around neurons in ganglia
Peripheral Nervous System
Central Nervous System
Ependymal cells
• Line ventricles (brain) and central canal (spinal cord)
• Assist in producing, circulating, and monitoring of cerebrospinal fluid
Schwann cells
• Surround axons in PNS
• Are responsible for myelination of peripheral axons
• Participate in repair process after injury
Oligodendrocytes
• Myelinate CNS axons
• Provide structural framework
Astrocytes
• Maintain blood-brain barrier
• Provide structural support
• Regulate ion, nutrient, and dissolved gas concentrations
• Absorb and recycle neurotransmitters
• Form scar tissue after injury
Microglia
• Remove cell debris, wastes, and pathogens by phagocytosis

NEURON CELL BODY LOCATION

- Most are found in the central nervous system
  - Gray matter – cell bodies and unmyelinated fibers
  - Nuclei – clusters of cell bodies within the white matter of the central nervous system
- Ganglia – collections of cell bodies outside the central nervous system

- ***WHITE MATTER:***

Bundles of nerve fibres and axons.

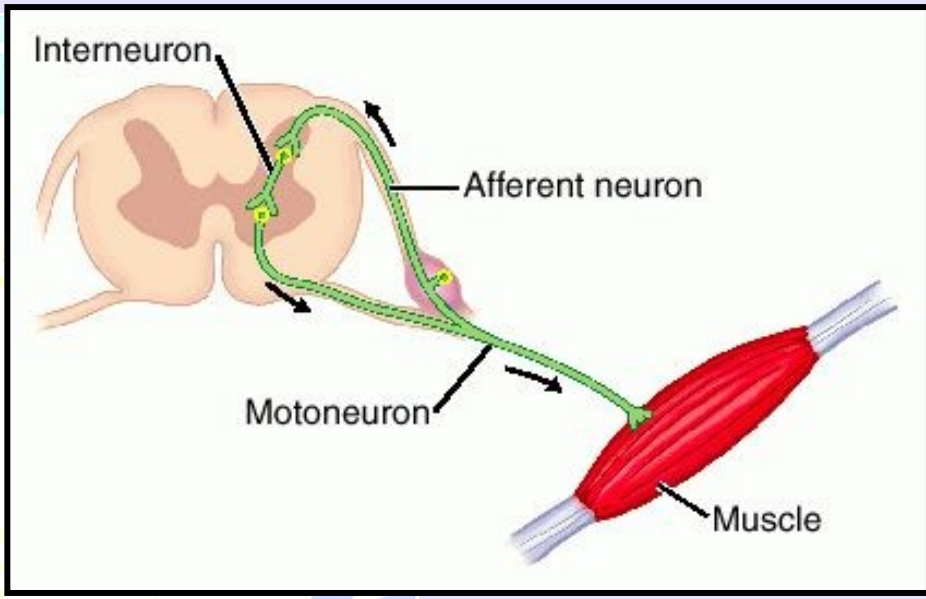
- Outside CNS (in PNS)-------NERVES
- Within CNS---------TRACTS

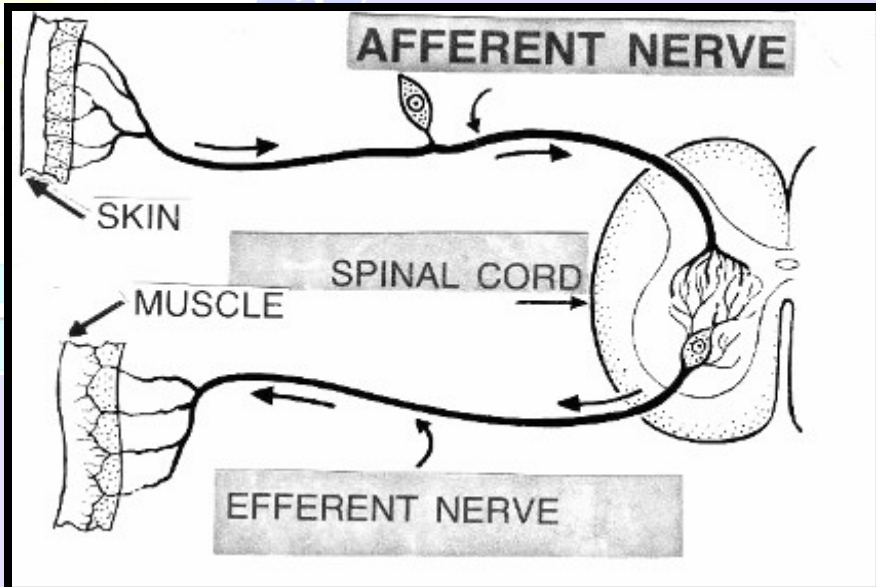
Functional classification of neurons

- **Afferent neurons:** convey information from tissues and organs to the central nervous system & are called **Sensory neurons**
- **Efferent neurons**:

transmit signals from the central nervous system to the effector cells and called **motor neurons**.

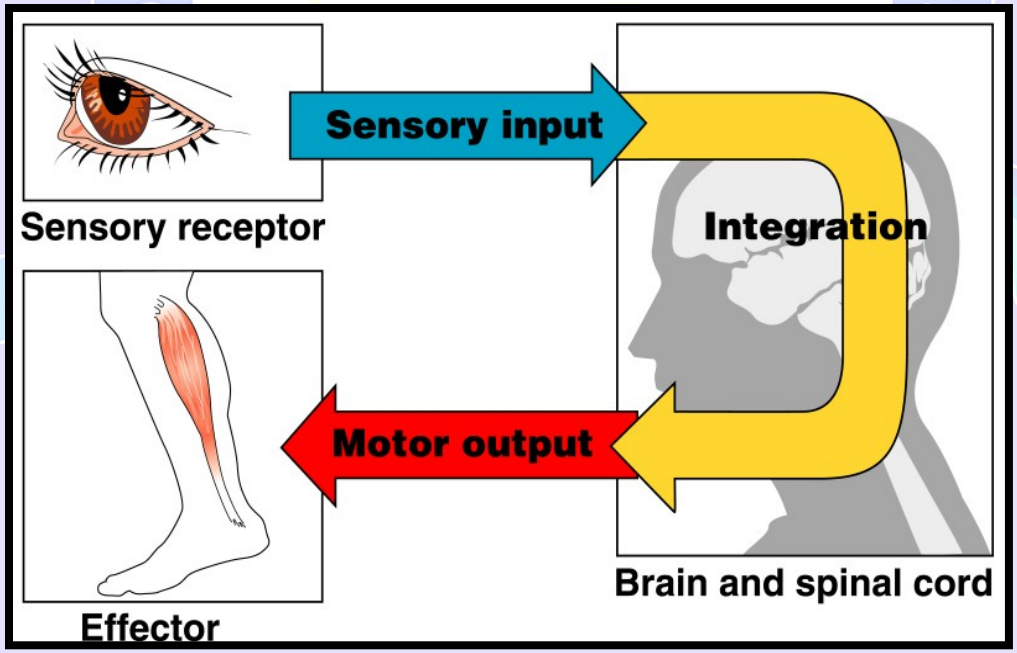
- **Interneurons** connect neurons within specific regions of the central nervous system.





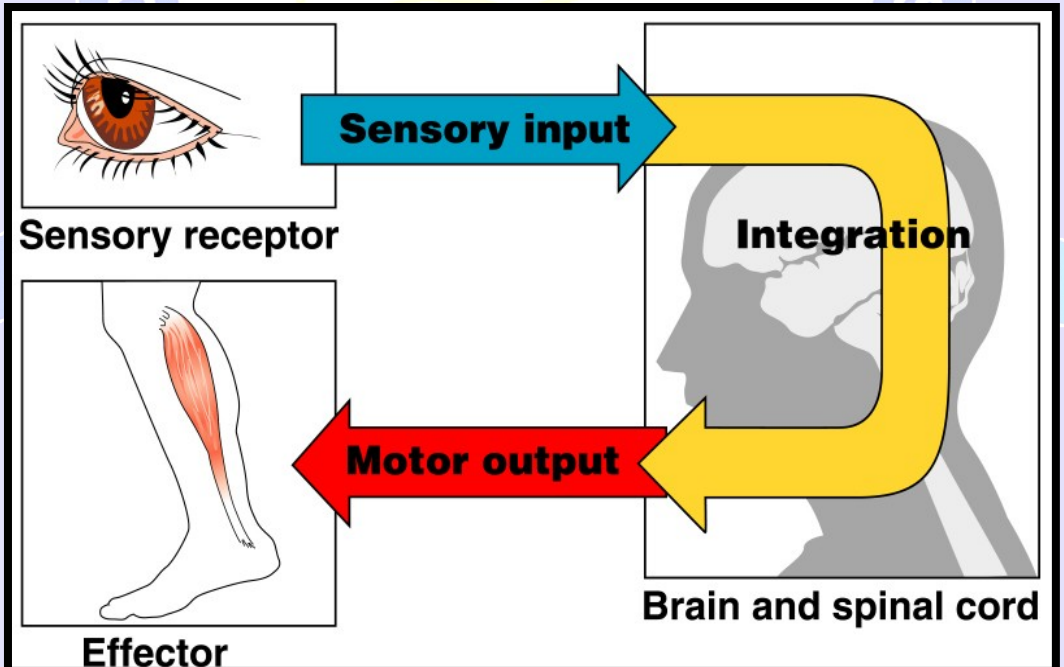
FUNCTIONAL CLASSIFICATION OF THE PERIPHERAL NERVOUS SYSTEM

- **Sensory (afferent) division**
  - Nerve fibers that carry information *to* the central nervous system



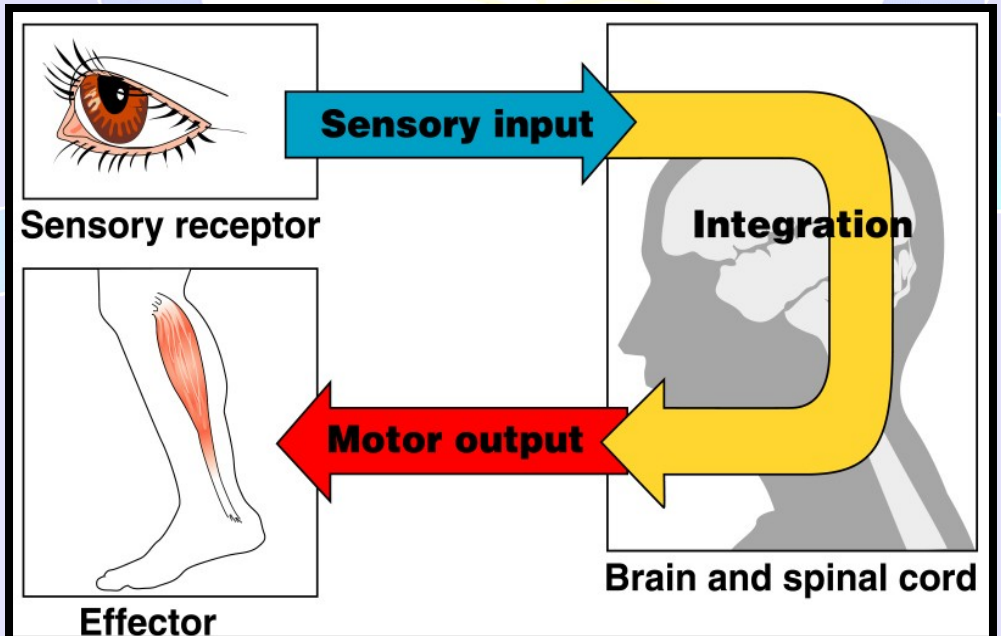
FUNCTIONAL CLASSIFICATION OF THE PERIPHERAL NERVOUS SYSTEM

- **Motor (efferent) division**
  - Nerve fibers that carry impulses *away from* the central nervous system



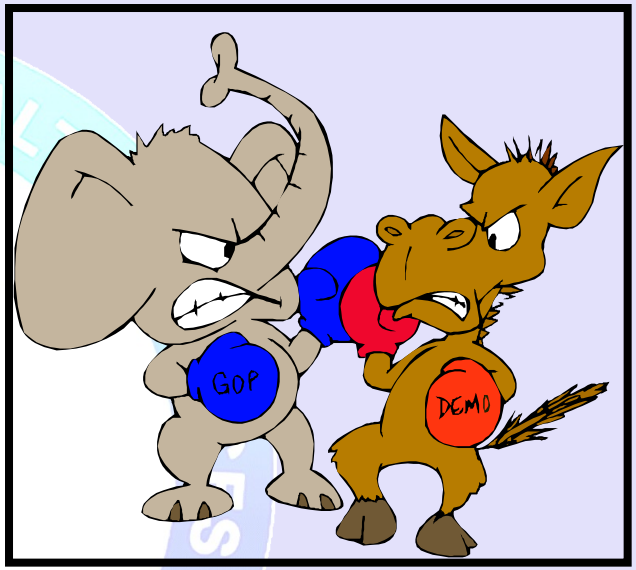
FUNCTIONAL CLASSIFICATION OF THE PERIPHERAL NERVOUS SYSTEM

- **Motor (efferent) division**
  - **Two subdivisions:**
    - Somatic nervous system = voluntary
    - Autonomic nervous system = involuntary



Autonomic Nervous System

- **2 divisions:**
  - **Sympathetic**
    - "Fight or flight"
    - "E" division
      - **Exercise, excitement, emergency, and embarrassment**
  - **Parasympathetic**
    - "Rest and digest"
    - "D" division
      - **Digestion, defecation, and diuresis**





Somatic vs. Autonomic

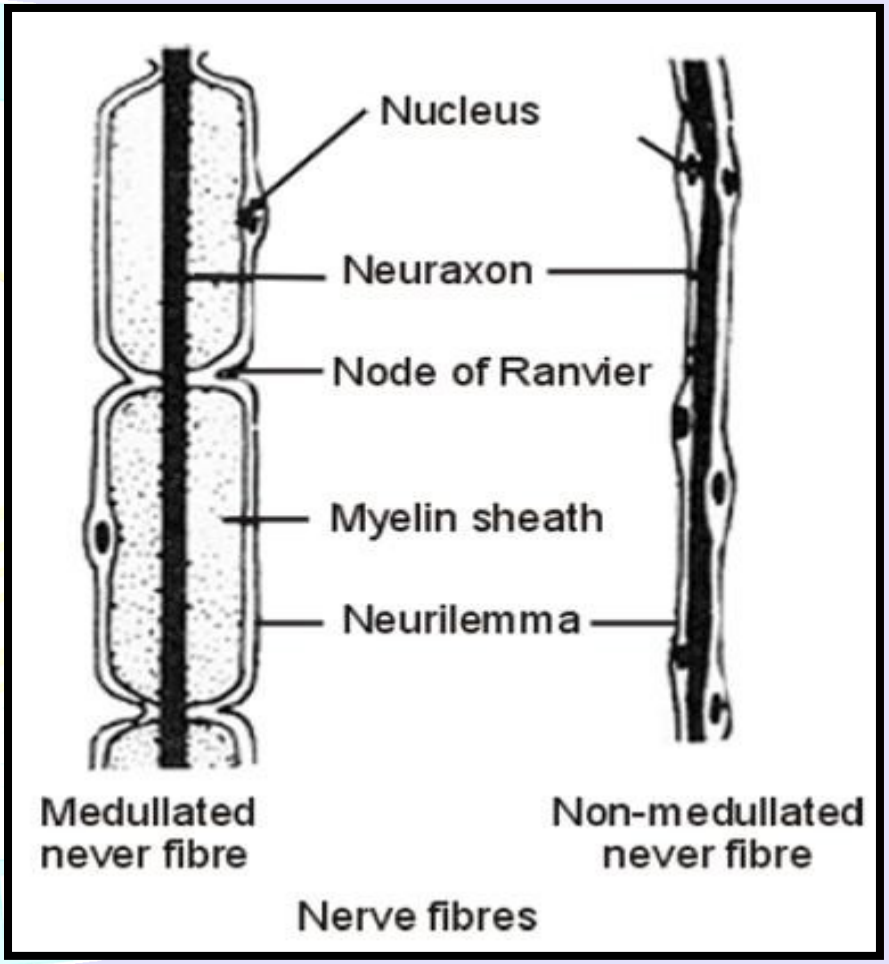
- Voluntary
- Skeletal muscle
- Single efferent neuron
- Axon terminals release acetylcholine
- Always excitatory
- Controlled by the cerebrum

- Involuntary
- Smooth, cardiac muscle; glands
- Multiple efferent neurons
- Axon terminals release acetylcholine or norepinephrine
- Can be excitatory or inhibitory
- Controlled by the homeostatic centers in the brain – pons, hypothalamus, medulla oblongata

NERVE FIBRE

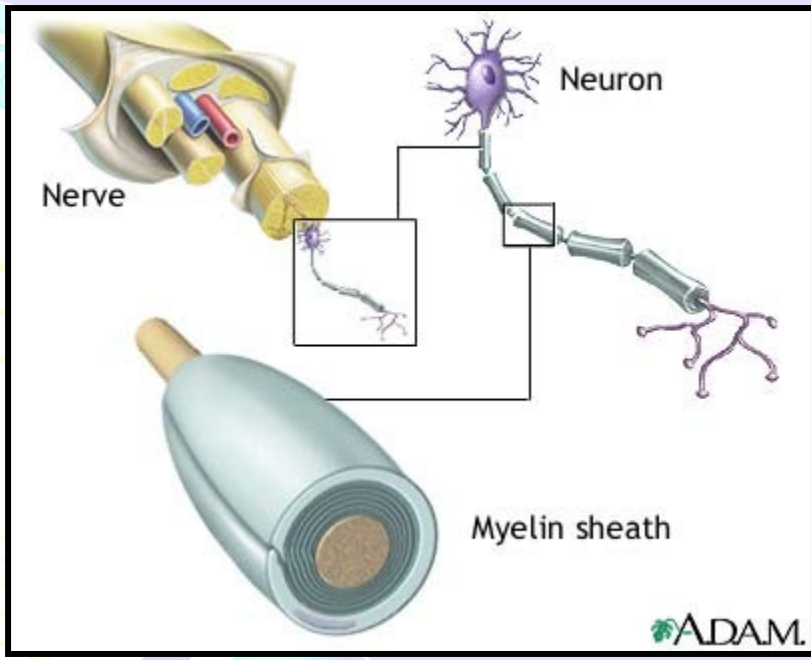
- Peripheral nerves
- White substance of the brain and spinal cord

- They are of two kinds:
  - Myelinated
  - Non-meylinated



MYELINATION

- To minimize metabolic expense while maintaining rapid conduction, many neurons have insulating sheaths of **myelin** around their axons.
- The sheaths are formed by **glial** cells:
- **Oligodendrocytes** in the central nervous system
- **Schwann cells** in the peripheral nervous system.

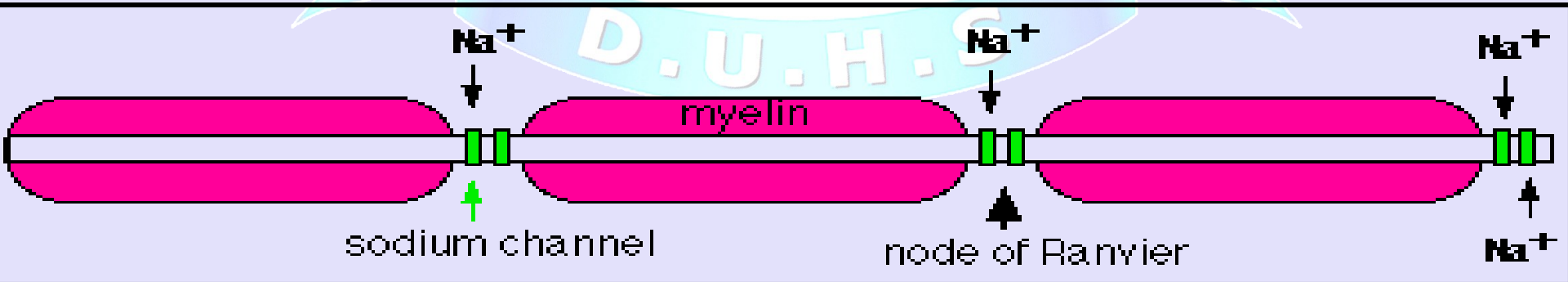




MYELINATION

- The sheath enables action potentials to travel **faster** than in unmyelinated axons of the same diameter, whilst using less energy.

- The myelin sheath in peripheral nerves normally runs along the axon punctuated by unsheathed **nodes of Ranvier** which contain a high density of voltage-gated ion channels.



REFERENCES

- *Keith L. Moore, Clinically oriented Anatomy,6th edition Page no: 46-49,53-56*


Thank You!